NEWTON SCHOOL

MATH | ANSWER KEY

Area of Shapes Homework — Answer Key

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Standard: *CCSS.MATH.CONTENT.6.G.A.1 — Find areas of triangles, special quadrilaterals, and polygons.*

Part A — Basic Area Practice

Problem #1 — Trapezoid

> A trapezoid has a top base of 0.6 in, a bottom base of 2 in, and a height of 1.7 in. Find the area.

Identify | Diagram | Pick Formula | Plug In | Calculate | Units

Step 1 — Identify. Given: $b_1 = 0.6$ in, $b_2 = 2$ in, $h = 1.7$ in. Find: Area A.

Step 2 — Draw. Trapezoid with top base $b_1 = 0.6$, bottom base $b_2 = 2$, height $h = 1.7$ (perpendicular distance between the two parallel bases).

Step 3 — Formula. $A = \dfrac{b_1 + b_2}{2} \cdot h$

Step 4 — Plug in. $A = \dfrac{0.6 + 2}{2} \cdot 1.7$

Step 5 — Calculate. $A = \dfrac{2.6}{2} \cdot 1.7 = 1.3 \cdot 1.7 = 2.21$

Step 6 — Final answer (with units!).

A = 2.21 in²

Problem #2 — Rectangle

> A rectangle has a width of 2.3 ft and a length of 5 ft. Find the area.

Identify | Diagram | Pick Formula | Plug In | Calculate | Units

Step 1 — Identify. Given: $w = 2.3$ ft, $\ell = 5$ ft. Find: Area A.

Step 2 — Draw. Rectangle: long side $\ell = 5$ ft, short side $w = 2.3$ ft.

Step 3 — Formula. $A = w \cdot \ell$ (rectangle: just multiply the two side lengths)

Step 4 — Plug in. $A = 2.3 \cdot 5$

Step 5 — Calculate. $A = 2.3 \cdot 5 = 11.5$

Step 6 — Final answer (with units!).

A = 11.5 ft²

Problem #3 — Trapezoid

> A trapezoid has a top base of 4.6 m, a bottom base of 1.6 m, and a height of 3.4 m. Find the area.

Identify | Diagram | Pick Formula | Plug In | Calculate | Units

Step 1 — Identify. Given: $b_1 = 4.6$ m, $b_2 = 1.6$ m, $h = 3.4$ m. Find: Area A.

Step 2 — Draw. Trapezoid with parallel bases $b_1 = 4.6$ (top) and $b_2 = 1.6$ (bottom), perpendicular height $h = 3.4$.

Step 3 — Formula. $A = \frac{b_1 + b_2}{2} \cdot h$

Step 4 — Plug in. $A = \frac{4.6 + 1.6}{2} \cdot 3.4$

Step 5 — Calculate. $A = \frac{6.2}{2} \cdot 3.4 = 3.1 \cdot 3.4 = 10.54$

Step 6 — Final answer (with units!).

A = 10.54 m²

Problem #4 — Triangle

> A triangle has a base of 4 m and a height of 2.7 m. Find the area.

Identify | Diagram | Pick Formula | Plug In | Calculate | Units

Step 1 — Identify. Given: $b = 4$ m, $h = 2.7$ m. Find: Area A.

Step 2 — Draw. Triangle with base $b = 4$ along the bottom, height $h = 2.7$ drawn perpendicular from the apex down to the base line.

Step 3 — Formula. $A = \frac{1}{2} \cdot b \cdot h$ (triangle is half of a rectangle)

Step 4 — Plug in. $A = \frac{1}{2} \cdot 4 \cdot 2.7$

Step 5 — Calculate. $A = \frac{1}{2} \cdot 4 \cdot 2.7 = 2 \cdot 2.7 = 5.4$

Step 6 — Final answer (with units!).

$A = 5.4\ m^2$

Part B — Apply Your Skills

Problem #5 — Trapezoidal Garden (application)

> A trapezoidal garden has area 96 cm^2 and height 6 cm. The longer base is 4 cm longer than the shorter base. Find both bases (b_1 and b_2).

Identify | Diagram | Pick Formula | Plug In | Calculate | Units

Step 1 — Identify. Given: $A = 96$ cm^2, $h = 6$ cm, and $b_2 = b_1 + 4$ (longer base is 4 cm more than the shorter). Find: b_1 and b_2.

Step 2 — Draw. Trapezoid: top (shorter) base labeled b_1, bottom (longer) base labeled $b_2 = b_1 + 4$, perpendicular height $h = 6$.

Step 3 — Formula. $A = \frac{b_1 + b_2}{2} \cdot h$

Step 4 — Plug in. $96 = \frac{b_1 + (b_1 + 4)}{2} \cdot 6 = \frac{2b_1 + 4}{2} \cdot 6 = (b_1 + 2) \cdot 6$

Step 5 — Solve. $\frac{96}{6} = b_1 + 2 \;\Rightarrow\; 16 = b_1 + 2 \;\Rightarrow\; b_1 = 14$

Then: $b_2 = b_1 + 4 = 14 + 4 = 18$.

Step 6 — Final answer (with units!).

b_1 = 14 cm (shorter), b_2 = 18 cm (longer)

Check: $A = \frac{14 + 18}{2} \cdot 6 = \frac{32}{2} \cdot 6 = 16 \cdot 6 = 96$ ✓

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